

Midterm Exam (Operating System, 2021)

答案卷請註明學號, 班別, 組別, 姓名, 以方便成績輸入. 滿分:100

1. (2%) 班別: [上午A班 | 下午B班], 組別:
2. (10%) Identify one advantage and one disadvantage of open-source operating system. Include the types of people who would find each aspect to be an advantage or a disadvantage. 舉出一個有利點, 一個不利點, 說明那些人會獲利或受損
3. (20%) The following tasks are being scheduled using a preemptive, round-robin scheduling algorithm. Each task is assigned a numerical priority, with a higher number indicating a higher relative priority. In addition to the tasks listed below, the system also has an idle task (which consumes no CPU resources and is identified as Thread *N*). The idle task has priority 0 and is scheduled whenever the system has no other available task to run. The length of a time quantum is 10 units. If a task is preempted by a higher-priority task, the preempted task is placed at the end of the queue with a fresh time quantum (i.e., 10 units). Please (a) draw the Gantt chart, and calculate (b) the CPU utilization for the first 100 time units, and (b) the turnaround time and waiting time of Process E. 被打斷的低優先thread會放置在隊伍尾端, 但是重新給予10單位時間, 計算前100單位時間的CPU利用率, Thread E的數據

Thread	Priority	Burst	Arrival
<i>N</i>	0	infinity	0
<i>A</i>	20	20	0
<i>B</i>	30	9	8
<i>C</i>	20	23	15
<i>D</i>	35	15	60
<i>E</i>	10	30	64
<i>F</i>	10	17	78

(time, running task-remaining time, ready queue) = (0, A20, empty), (8, B9, A12), (15, B2, A12-C23), (17, A12, C23), (27, C23, A2), (37, A2, C13), (39, C13, empty), (49, C3, empty), (52, N, empty), (60, D15, empty), (64, D11, E30), (70, E30, D5), (78, E22, D5-F17), (80, D5, F17-E20), (85, F17, E20), (95, E20, F7), (105, F7, E10), (112, E10, empty), (122, N, empty),
 $(100-8)/100=92\%$,

turnaround time: $122-60=62$, waiting time: $62-30=32$

4. (10%) (a) Please list the four evaluation (評估) methods for CPU scheduling. (b) Which one of them is often used in classroom (教室中常用哪一種)?
5. (10%) Please compare “multiprocessor systems” to “Clustered Systems”. 比較兩者
6. (10%) Please give an example to show that multi-process approach is more efficient than single-process for uni-CPU, uni-core system. 請舉出一個實例說明, 既使只有一個CPU, 一個core, 採取multi-process的方式會比較(相對於single process)有效率。
7. (10%) SJF scheduling有兩種版本, preemptive 以及non-preemptive的, FCFS是否也有兩種版本? **NO**, Round-Robin scheduling是否也有兩種版本? **YES** 說明理由。FCFS以到達時間為優先序, 晚到的priority 較低, 永遠不會

preemptive, RR有兩種版本, 尚未逾時的行程是否會被高優先的行程踢走

8. (8%) What is the fundamental knowledge of a Junior student of Computer Science in “the usage of thread”? 資工系大三學生對於”thread的使用”應該具備那些知識?
9. (20%) Explain the following terms. Please demonstrate your knowledge in associate with those terms, instead of translations. Examples are very welcome. Clear description helps credit. (a) programmable interrupt controller (b)scheduler and dispatcher (c)thread control block (d) dual mode
解釋下列名詞。請展現對於作業系統的理解程度, 切勿翻譯了事。盡可能多用例子。字數太少, 資訊不足都很可能導致無法正確的評估。因此, 請盡量多寫, 寫清楚。惜墨如金的, 責任自負。